

SHOULD HAVE A BETTER TITLE

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ABSTRACT

We proposed a novel paradigm for implementing latent reasoning by combining DAPO-like reinforcement learning and special gating units. In this way, we transformed text-based explicit reasoning models to latent reasoning models, progressively and practically. Initial experiments demonstrate its feasibility, with significant stability and the least extra training overhead by parallelism and minimal changes, while holding full potential to scale up.

1 INTRODUCTION

Waiting for rewrite.

2 RELATED WORK

Explicit Reasoning Models and Large-scale Reinforcement Learning Advances in *explicit* reasoning models, including OpenAI o1 & o3 (OpenAI, 2025; 2024) and DeepSeek R1 (DeepSeek-AI, 2025), demonstrated their strong ability in mathematical reasoning. Their work mainly focused on scaling up test-time computation by generating long Chain of Thought (CoT) (Wei et al., 2022) text sequences including symbolic inference and reflection.

Advances in these models can’t leave out the strong power of large-scale reinforcement learning paradigm, which has been widely used in LLMs. Ouyang et al. (2022) showed how GPT-3 (Kojima et al., 2023) can be fine-tuned with reinforcement learning via Proximal Policy Optimization (PPO) Schulman et al. (2017), which optimizes the agent’s policy by maximizing the expected reward while ensuring the new policy does not deviate excessively from the old policy. Shao et al. (2024) Proposed Group Relative Policy Optimization (GRPO), which overcame the limitations of PPO, including the need for a large number of samples and the difficulty of tuning hyperparameters. Yu et al. (2025) Demonstrated the effectiveness of Decoupled Clip and Dynamic Sampling Policy Optimization (DAPO) in large-scale reinforcement learning, which performed best in several benchmarks on Qwen2.5-32B Yang et al. (2024b) solely based on pure reinforcement learning and the base instruction model of Qwen2.5-32B (Yang et al., 2024b).

In contrast, our work leverages a latent space instead of explicit reasoning text, which reduces verbosity and improves efficiency. Based on large-scale reinforcement learning, we effectively transplant those explicit reasoning-based optimization and advances into the latent reasoning region.

Latent Space Reasoning Unlike explicit reasoning, latent reasoning paradigms leverage latent spaces, promoting efficiency and accuracy. Previous works have already found that LLMs may involve hidden computation steps, which are not explicitly generated in text (Yang et al., 2024c; Chen et al., 2023). Driven by this spirit, Hao et al. (2024) proposed COCONUT, assigning models to “reason in a latent continuous space” through specifically designed training algorithms, and demonstrated its potential compared to the explicit Chain-of-Thought reasoning procedure, especially on tasks requiring backtracking.

Geiping et al. (2025) Introduced a new recurrent transformer architecture that lets models perform multiple hidden reasoning iterations per token, as they suggested that “a recurrent block can be looped arbitrarily at inference to refine the hidden state before producing the next token.” Saunshi et al. (2025) Demonstrated that the critical impact of model capabilities lies in its depth instead of parameter size. In this way, they proposed *Looped Transformer*, which generated latent thoughts

and simulated Chain-of-Thought reasoning procedures internally. They also pretrained a 3.5B model that might compete with 50B ones.

In contrast, our work focuses on the practical implementation of latent reasoning, instead of cold-start training, which is done by Saunshi et al.. Our paradigm for feasibility and stability is based on the approach of training the model without specifically designed trajectories, such as stepping guidance, from a production-level model via small injections with VAE and gates. It also combines the advances of explicit reasoning models and large-scale reinforcement learning objectives. At the same time, prior work mainly focused on producing the latent reasoning space, and small-scale usages primarily based on SFT.

3 METHODOLOGY

We first introduce the general structure of modern large language models (LLMs) to demonstrate our approach and paradigm better.

Modern LLMs involve a Embedding layer; a core Transformer, including Self-Attention and Feed-Forward Network (FFN); and the Output linear layer which projects the latent state vector of `embed_dim` to probability distribution of `vocab_size` layers, as we denote them E , M_n and O respectively, which n indicates the layer of hidden state (Vaswani et al., 2017; Radford et al., 2019).

3.1 MODEL ARCHITECTURE

We used layers in the middle of the Transformer core to enter the latent reasoning space to avoid breaking special usage in final layers (Saunshi et al., 2025). Specifically, the latent vector we chose to extract activates the n_0 -th decoder layer within the n -layer decoder.

We preliminarily perform prefilling by the question with the length q_0 into the model and get the prefilled KV Cache (Pope et al., 2022) and the probability distribution of the answer’s first token t_{q_0+1} (starting to count from t_1).

After extracting the hidden state h_{n_0, q_0} (where n_0 denotes the layer number of the Transformer and q_0 denotes question length), we deliver the state to a Variational Autoencoder (Kingma and Welling, 2019), and produce a vector for storage with length dimension of `compressed_dim`.

3.1.1 VARIATIONAL AUTOENCODER

We use Variational Autoencoder (VAE) (Kingma and Welling, 2019) to compress and uncompress the hidden state token $h_{n,k}$, which is an l_h -dimensional vector, to an l_c -dimensional vector and then back to l_h . The VAE layer looks like a bottleneck, compressing the hidden state to a lower dimension and then expanding it back to its original size. The shape of the VAE layer in our work can be described in Figure 1.

We pre-trained the VAE on the online-generated activation of 1024 consequences, and the validation loss decreased steadily, proving that the VAE’s pre-training process requires little data. In the appendix, we provide supplemental data to demonstrate the effectiveness of this approach.

Long-term Cache We will store the latent space activation output from later n_0 for further parallel training. Yet directly storing the raw vector of `embed_dim` could be space-consuming. (See ¹ for quantitative analysis of the space consumption, and we have provided data in B.1.)

We store it in each operation and concatenate these vectors into a matrix subsequently as a hidden state.

3.1.2 GATES

Our model introduces the update gate and the Dynamic Current Step Mixing Gate.

¹Consider a sample batch size of 1024, sequence length 1024, and embedding dimension 1536. Each element occupies 2 bytes (BF16), resulting in $1024 \times 1024 \times 1536 \times 2 = 2^{31.5}$ bytes, or approximately 2.8 GB of VRAM. In contrast, if we adapt the VAE layer to compress the hidden state to 256 dimensions, we only need $1024 \times 1024 \times 256 \times 2 = 2^{29}$ bytes, or approximately 512 MB of VRAM.

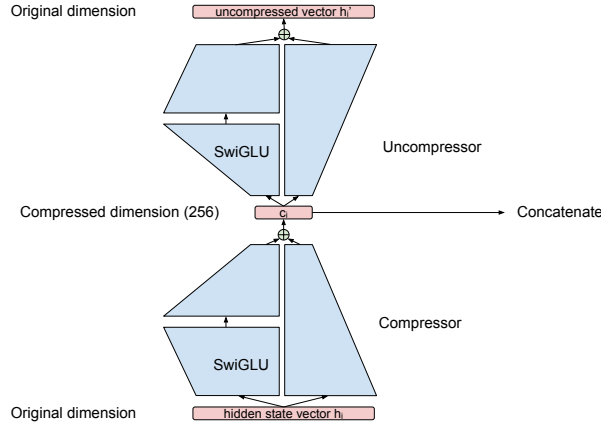


Figure 1: The VAE layer. We first compress the hidden state to a lower dimension (expand into 4 times the original size and then compress), extracting the compressed vector, and use the symmetric approach to expand it back to the original dimension.

1. **Update Gate.** Inspired by Gated Recurrent Unit (GRU) (Zhang et al., 2023; Chung et al., 2014; Cho et al., 2014), the update gate is used to increase its contextual consistency in patent space, but the value of the gate is derived from the sequence. We denote it as g_t , and its derivation procedure can be described in Equation (not yet). We initialize it to all zeros, i.e., initially the model is equivalent to the original one, but it evolves as g moves away from 0 .
2. **Dynamic Current-Step-Mixing Gate (DMG).** The injection ratio of the latent state vector needs to vary depending on the next token, yet directly applying one fixed gate for all cannot fit this need. Therefore, we add a DMG to gate the Update Gate according to both the latent state vector and the embedding of the next token. We fill the DMG with full zero initializations, which aims to stabilize the training process. The DMG is composed of a linear layer and a sigmoid activation function. The linear layer projects the concatenated vector of the hidden state vector and the embedding vector to a vector half the size of the original.

The processed vector $h'_{n,k}$ after the VAE layer is passed to several gates, as we denote in Equation 1.

$$h'_{n,k} = h'_{n,k-1} \odot g \odot \text{DMG}(e_k) \times s_{\text{inj}} + (1 - g) \odot e_k \quad (1)$$

Where $e_k = \text{Embedding}(t_k)$; t_k denotes the token of the sequence with position k , and s_{inj} denotes the injection scalar, which is a constant scalar to bridge the difference in order of magnitude between the gating value and the result of Dynamic Current-Step-Mixing Gate, which will be described in Section 3.1.2.

3.2 INFERENCE

We describe the inference process incorporating latent state injection and VAE-based compression in Algorithm 1. At each generation step, the model extracts the current hidden state, compresses it via a Variational Autoencoder (VAE), and stores the latent representation in a cache. The latest latent code is then decompressed and optionally refined through reasoning layers before being used to compute the logits for the next token. This process iteratively updates the attention context and position encoding, enabling latent-informed token generation conditioned on the evolving latent memory.

Algorithm 1: Inference with VAE and Latent Injection

Input: Tokenized input $t_{1:q_0}$, attention mask m
Output: Sampled tokens $t_{q_0+1:q}$, log-probabilities, latent cache

- 1 Initialize position index p and KV Cache;
- 2 **if** $depth > 0$ **then**
- 3 Initialize deep cache stack
- 4 Embed input and perform model forward to extract h_{n_0, q_0} ;
- 5 Initialize latent cache $z \leftarrow \mathbf{0}$ and output buffer $t \leftarrow \mathbf{0}$;
- 6 **for** $i \leftarrow 1$ **to** max_length **do**
- 7 Compress latest hidden h_{n_0, q_0+i-1} via VAE and append to z ;
- 8 Uncompress z_{-1} via VAE to obtain $\tilde{h}$;
- 9 **if** $depth > 0$ **then**
- 10 Loop $\tilde{h}$ through d reasoning layers to get updated $\tilde{h}$
- 11 Compute next-token logits $\ell \leftarrow \text{LMHead}(\tilde{h})$;
- 12 Sample token t_{q_0+i} from $\text{TopK}(\text{Softmax}(\ell/T), k)$;
- 13 Append t_{q_0+i} and log-prob to result;
- 14 **if** *eot token appears in all sequences* **then**
- 15 break
- 16 Update attention mask and position;
- 17 Embed t_{q_0+i} and update hidden via model forward with latent injection
- 18 **return** *Sampled tokens, log-probs, latent cache*

3.3 TRAINING

It is hard for us to train models in a budget-tight scenario, so we adapted Low-Rank Adaptation (LoRA) (Hu et al., 2021) to our model. We utilize the DAPO algorithm as our reinforcement learning framework; however, the entire procedure differs slightly from the original DAPO. The whole training procedure is described in Algorithm 2.

3.3.1 REINFORCEMENT LEARNING

Training Objective We adapt the DAPO (Yu et al., 2025) algorithm, but we have not implemented the clip mechanism, which is used to clip the reward and penalty to a specific range. Hence, we currently use the original DAPO algorithm, which is shown in Equation 2. We parallelized several inference procedures simultaneously, preferably 16 for the same question, and verified the answer (Kydliček).

$$\begin{aligned}
 \mathcal{J}_{\text{DAPO}}(\theta) = & \mathbb{E}_{(q,a) \sim \mathcal{D}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \\
 & \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} r_{i,t}(\theta) \hat{A}_{i,t} \\
 \text{subject to } & 0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G,
 \end{aligned} \tag{2}$$

Where

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}, \quad \hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}. \tag{3}$$

Award Mechanism We use $\text{reward}_{\text{acc}}$ and $\text{penalty}_{\text{len}}$ to award or penalize in accuracy, according to DAPO (Yu et al., 2025). The exact value of the reward and penalty will be described in the appendices.

Positive-Negative Clipping We choose only all the positive samples to eliminate the issue of too many negative samples (each of which is less useful for improving the model) under a budget-tight scenario. We randomly select the remaining negative samples in a ratio of 1 : 1.

Algorithm 2: Training Procedure**Input:** Training data (x, y) , sampling model, verifier, parameters**Output:** Trained model

```

1 Initialize  $step \leftarrow 0$ ;
2 while  $step \leq total\_steps$  do
3   foreach  $(problems, ans)$  in  $data\_train$  do
4     Tokenize problems to  $input\_ids$  and  $attn\_mask$ ;
5     Call  $sampler()$  and update intermediate results, shown in Algorithm 1;
6     Compute  $correctness\_rewards \leftarrow verifier()(res, ans)$ ;
7     Compute  $len\_rewards$  from  $text\_end\_indices$ ;
8     Compute  $correctness\_rate$ ;
9     if  $correct\_samples$  too few then
10       Downsample incorrect results to balance;
11     Normalize reward: clip too short/long sequences, compute total reward;
12     Build final training sequence by concatenating  $input\_ids$  and  $res$ ;
13     for  $epoch \leftarrow 1$  to  $num\_epochs$  do
14       for  $i \leftarrow 0$  to  $N$   $step\_batch\_size$  do
15         Extract a batch of sequence and hidden cache;
16         Decode latent via VAE and optionally apply recursive looping;
17         Combine token and latent via Gater to produce training embedding;
18         Run model forward to get logits;
19         Compute  $loss$  by mixing DAPO loss and gating value bonus;
20         Update latent cache randomly for a subset;
21         Backpropagate and optimize;
22 return  $trained\_model$ 

```

3.3.2 TIME CONSISTENCY REGULARIZATION

We adopt a mechanism called the ‘‘Gating Value Bonus,’’ which allows the model to fully leverage the information from the latent space instead of getting stuck in the local optima of not using the space sufficiently.

The gating value bonus is a regularization term designed to constrain gating values toward the midpoint, as the formulation is shown in Equation 4.

$$\mathcal{J}_{\text{gate}}(\mathbf{g}) = \frac{1}{\text{output_dim}} C_0 \sum_{i=0}^{\text{output_dim}} \exp \left[\lambda \left(\frac{1}{2} - g_i \right)^2 \right] \quad (4)$$

Where $g_i = \mathbf{g} \cdot \mathbf{e}^{(i)}$, i.e., the i -th element of the gate vector $\mathbf{g}$, and C_0 is a constant to control the scale of the bonus; λ is a constant to control the rate of convergence.

4 EXPERIMENTS

4.1 SETUP

Base Model We use Qwen3-4B (Yang et al., 2024a;b; Team, 2025) as our base model, which is a 4B-parameter model with 24 layers. DeepSeek distilled the model into an explicit reasoning model (DeepSeek-AI, 2025), which is our comparison baseline.

Baselines We compare our model with the following baselines:

- **Qwen3-1.7B:** The original model.
- Distillation method from DeepSeek-AI (2025) performed on Qwen3-4B.

- The original model with DAPO (Yu et al., 2025) objective solely.
- Our model with DAPO algorithm, gating mechanism, and VAE layer.
- The original model with VAPO (Yue et al., 2025) objective solely.

5 DISCUSSION

After conducting several experiments and analyzing the theoretical bases, we have several discussions:

6 CONCLUSION

In this paper, we proposed...

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A DETAILED METHODOLOGY

A.1 DETAILED FORWARD PROPAGATION

In this section, we will provide a detailed description of the forward propagation process in our proposed model. The procedure of forward propagation is shown in Algorithm 3.

Algorithm 3: MODELFORWARD Procedure

Input: Hidden state h , attention mask m , position p , KV cache k (optional), injected hidden state $\tilde{h}$ (optional), injection function f_{inject} (optional), start layer l_s , end layer l_e , extraction index l_x (optional)

Output: Final hidden state, optionally intermediate state at layer l_x

```

1 Compute causal mask  $m_c \leftarrow \text{UPDATECAUSALMASK}(m, h, p, k)$ ;
2 Compute positional embedding  $e_p \leftarrow \text{ROTARYEMBED}(h, p)$ ;
3 for  $i \leftarrow l_s$  to  $l_e$  do
4   if  $\tilde{h} \neq \emptyset$  and  $i = l_{\text{inject}}$  then
5     if  $f_{\text{inject}} = \emptyset$  then
6        $h \leftarrow \text{GATER}(\tilde{h}, h)$ ;
7     else
8        $h \leftarrow f_{\text{inject}}(\tilde{h}, h)$ ;
9    $h \leftarrow \text{TRANSFORMERLAYER}_i(h, m_c, p, k, e_p)$ ;
10  if  $i = l_x$  then
11     $h_{\text{spec}} \leftarrow h$ ;
12 if  $l_x \neq \emptyset$  then return  $h, h_{\text{spec}}$ ;
13 else return  $h$ ;
```

B DETAILED EXPERIMENTS

B.1 VARIATIONAL AUTOENCODER LAYER

We set the VAE layer with parameters 2048, 256, and 2048×4 , and trained it on a specifically tidied Open Thought dataset. The learning rate is set to 1×10^{-3} , using the Adam optimizer (Kingma and Ba, 2014) with batch size 20. The training process and loss in 15 epochs are shown in Figure 2.

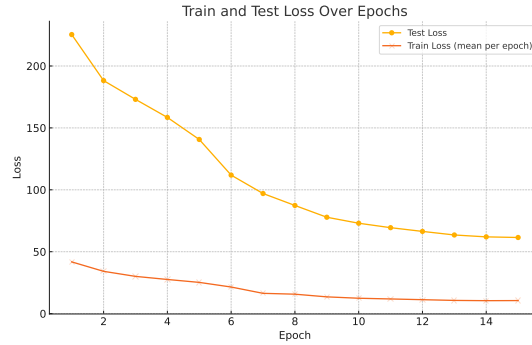


Figure 2: The training process of the VAE layer. The reconstruction error of the hidden state vector calculates the loss. The training process is stable and converges quickly.